

Foundation (Jalapeno)

- Q1.** The digit 7 in the number 4732 represents:
(A) Thousand
(B) Hundred
(C) Ten
(D) One
(E) Ten Thousand
- Q2.** A space station measures 4321 meters in length. Rounded to the nearest thousand, the length is:
(A) 4000
(B) 5000
(C) 3000
(D) 2000
(E) 1000
- Q3.** The number 943 in expanded form is:
(A) $900 + 40 + 3$
(B) $900 + 40 + 4$
(C) $900 + 4 + 3$
(D) $900 + 43$
(E) $900 - 40 + 3$
- Q4.** A physicist counts 147 particles in an experiment. The digit 1 represents the number of:
(A) Ones
(B) Tens
(C) Hundreds
(D) Tens of thousands
(E) Thousands
- Q5.** 246 rounded to the nearest ten is:
(A) 240
(B) 250
(C) 260
(D) 200
(E) 300
- Q6.** The digit 5 in the number 3582 represents the place value of:
(A) Tens
(B) Ones
(C) Hundreds
(D) Thousands
(E) Ten Thousands
- Q7.** A planet has 7532 moons. The number 7532 in standard form is $7 \cdot 10^3 + 5 \cdot 10^2 + 3 \cdot 10^1 + 2 \cdot 10^0$
(A) $7 \cdot 10^3 + 5 \cdot 10^2 + 3 \cdot 10^1 + 2 \cdot 10^1$
(B) $7 \cdot 10^4 + 5 \cdot 10^3 + 3 \cdot 10^2 + 2 \cdot 10^1$
(C) $7 \cdot 10^3 + 5 \cdot 10^2 + 3 \cdot 10^1 + 2 \cdot 10^0$
(D) $7 \cdot 10^3 + 5 \cdot 10^2 + 3 \cdot 10^0 + 2 \cdot 10^1$
(E) $7 \cdot 10^2 + 5 \cdot 10^1 + 3 \cdot 10^0 + 2 \cdot 10^3$
- Q8.** The number 135 in expanded form is $1 \cdot 10^2 + 3 \cdot 10^1 + 5 \cdot 10^0$
(A) $1 \cdot 10^2 + 3 \cdot 10^1 + 5 \cdot 10^1$
(B) $1 \cdot 10^2 + 3 \cdot 10^0 + 5 \cdot 10^1$
(C) $1 \cdot 10^1 + 3 \cdot 10^2 + 5 \cdot 10^0$
(D) $1 \cdot 10^0 + 3 \cdot 10^1 + 5 \cdot 10^2$
(E) $1 \cdot 10^2 + 3 \cdot 10^1 + 5 \cdot 10^0$

Open Ended Questions (FRQ)

- Q9.** A laboratory contains 6274 units of a chemical. Write the number 6274 in expanded form and explain what each digit represents.
- Q10.** Round 9421 to the nearest hundred and explain the process used.

Chef's Tips

- Use real-life examples from space and science to illustrate place value concepts
- Emphasize the importance of precise calculation and attention to detail
- Encourage students to use diagrams or visual aids to support their understanding